Zeeshan Manzoor Bhat

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Zeeshan Manzoor Bhat
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Research Interests

Large-scale experimental testing; Nonlinear modelling of structures; Seismic hazard, risk and vulnerability assessment; Performance-based earthquake engineering; Resilience-based seismic design.

Education

Ph.D. in Structural and Earthquake Engineering *July 2018 – March 2025*
Indian Institute of Technology Roorkee, India
Thesis: *Seismic Performance of Masonry Infilled RC Frame Buildings Designed for Modern Codes*
Advisors: Prof. Yogendra Singh, Prof. Pankaj Agarwal

M.Tech. in Structural Engineering *July 2016 – July 2018*
National Institute of Technology Srinagar, India
Thesis: *An Experimental Investigation into the Behavior of Steel–Timber Composite Beams*
Advisor: Prof. Javed Ahmad Bhat

B.Tech. in Civil Engineering *July 2012 – June 2016*
National Institute of Technology Srinagar, India

Research Experience

Research Associate *June 2025 – Present*
Indian Institute of Technology Roorkee, India
Project: *Enhancing Sustainability of RC Frame Buildings through Seismic Safety and Design Life Elongation Using GFRP Reinforcement*

Funded Projects

Seismic Safety Measures for RC Frame Buildings with Different Types of Infill Panels
Funding: INR 38 Lakh, NBCC (India) Limited
Institution: Indian Institute of Technology Roorkee, India

Publications

[1] **Bhat, Z. M.**, & Singh, Y. (2026). Out-of-plane seismic fragility analysis of masonry infills in low- and mid-rise RC frame archetype buildings accounting for in-plane damage. ***Journal of Building Engineering***. DOI

[2] **Bhat, Z. M.**, & Singh, Y. (2025). Seismic risk assessment of fly ash brick and AAC block masonry infilled RC frame buildings designed for modern codes. ***Journal of Earthquake Engineering***. DOI

[3] **Bhat, Z. M.**, & Singh, Y. (2025). Out-of-plane behavior of masonry infills of different types and slenderness ratios under reversed cyclic loading. ***Journal of Structural Engineering***. DOI

[4] **Bhat, Z. M.**, Singh, Y., & Agarwal, P. (2024). Cyclic testing and diagonal strut modelling of different types of masonry infills in reinforced concrete frames designed for modern codes. ***Engineering Structures***. DOI

[5] **Bhat, Z. M.**, Singh, Y., & Agarwal, P. (2023). Characterization of mechanical behavior of different types of masonry with full-field strain using digital image correlation. ***Construction and Building Materials***. DOI

[6] Waseem, S. A., **Bhat, Z. M.**, & Bhat, J. A. (2022). Experimental investigation into the behavior of steel–timber composite beams. ***Practice Periodical on Structural Design and Construction***. DOI

Under Review
[7] **Bhat, Z. M.**, & Singh, Y. Cyclic behavior and damage assessment of masonry infilled RC frames with openings using DIC. ***Engineering Structures***

International Conferences

[1] **Bhat, Z. M.**, & Singh, Y. (2025). Reversed out-of-plane cyclic testing of masonry infills using double-leaf airbags. In ***COMPDYN 2025: Proceedings of the 10th International Conference on Computational Methods in Structural Dynamics and Earthquake Engineering***, Rhodes Island, Greece, 15–18 June 2025. DOI

[2] **Bhat, Z. M.**, & Singh, Y. (2025). Cyclic in-plane behavior of masonry infilled RC frames with detailed interpretation of damage using digital image correlation. In ***15th Canadian Masonry Symposium***, Ottawa, Canada, 2–5 June 2025.

Technical Reports

[1] **Bhat, Z. M.**, Singh, Y., & Agarwal, P. (2024). *Seismic Safety Measures for RC Frame Buildings with Different Types of Infill Panels.* NBCC (India) Limited, India.

[2] Team Member. (2022). *Compendium of Traditional Earthquake-Resilient Construction Practices.* NDMA, Government of India. Report Link

Open Datasets

[1] **Bhat, Z. M.**, & Singh, Y. (2024). Fly ash brick and AAC block masonry infilled RC frame buildings: OpenSees models and responses. ***DesignSafe-CI***. DOI

Awards and Achievements

- Canada Impact+ Research Training Award, Natural Sciences and Engineering Research Council of Canada (NSERC), 2026
- MHRD Fellowship for Doctoral Studies, Ministry of Human Resource Development, Government of India
- MHRD Fellowship for M.Tech Studies, Ministry of Human Resource Development, Government of India
- Ranked among the top 2% in the Graduate Aptitude Test in Engineering (GATE) 2016, among 119,873 candidates

Technical Skills

- **Programming & Tools:** Python, MATLAB, TCL, LaTeX
- **Software:** OpenSees, STKO, ETABS, SAP2000, OpenQuake
- **Experimental Techniques:** Digital Image Correlation (DIC), Laboratory Instrumentation and Data Acquisition, Servo-hydraulic Actuator Control, Load Cell Calibration

International Workshops

- **6th International Course on Seismic Analysis of Structures** 19–22 July 2021
  University of Palermo, Italy
- **Nonlinear Modelling of RC Structures Using OpenSees** 22–26 June 2020
  Università degli Studi di Napoli Federico II, Italy

Teaching Assistantships

- **Advanced Earthquake Resistant Design of Structures (EQN-512)** IIT Roorkee
  Instructor: Prof. Yogendra Singh
- **Earthquake Resistant Design of Structures (EQN-563)** IIT Roorkee
  Instructor: Prof. Yogendra Singh

References

Prof. Yogendra Singh
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Prof. Pradeep Bhargava
Professor, Department of Civil Engineering, Indian Institute of Technology Roorkee, India
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